

Section 3.3 Diagonalization and Eigenvalues

Exercise 3.3.3

Show that A has $\lambda = 0$ as an eigenvalue if and only if A is not invertible.

Start with $\det(A - \lambda I) = 0$

$\rightarrow$

Must be 0

as det of A

$\det(\text{matrix}) = 0 \Rightarrow ! \text{invert}$

Sub $\lambda = 0$ get

$\det(A) = 0;$

$\therefore \lambda = 0$ eigenvalue of A

IFF A !invertible.

Exercise 3.3.6

Find the characteristic polynomial of the $n \times n$ identity matrix I . Show that I has exactly one eigenvalue and find the eigenvectors.

where p is $\det(A - \lambda I)$ Then for square matrix I ;
we can define ~~$A = (\lambda I)$~~ $A = I$

$$\Rightarrow \det(I - \lambda I)$$

factor out 'I'

$$\Rightarrow (I(1 - \lambda))$$

Then we know there
only one eigenvalue $\lambda = 1$;

for eigen vector;

Because $(A - \lambda I) = 0$

has ∞ solutions;

The eigenvector = $\mathbb{R}^n$

where $n = n \times n$ matrix size

$$c_A(x) = \det(xI - A);$$

A char poly

Exercise 3.3.7

Given $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ show that: $\text{tr}(A)$? or $\text{tr}(Ax)$? ...

a. $c_A(x) = x^2 - \text{tr } Ax + \det A$, where $\text{tr}(A) = a + d$ is called the trace of A .

1) Find ~~$c_A(x)$~~ $= x^2 - ax + dx + ad - bc$

if $\text{tr}(A)x$

2) $\det(A) = ad - bc$; $\Rightarrow$

Then to find $xI - A \Rightarrow \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} x-a & -b \\ -c & x-d \end{bmatrix}$

Then finding det gives us $\det(xI - A)$ or $(x-a)(x-d) - (bc)$

which is simply $x^2 - dx - ax + ad - bc = x^2 - dx - ax + ad - bc$

or as grouped we have

$x^2 - (ad)x + (ad - bc) \Rightarrow x^2 - \text{tr}(A)x + \det(A) = c_A(x) \therefore \text{TRUE}$

$\uparrow$ $\uparrow$
 tr $\det$

Section 5.1 Subspaces and Spanning

Exercise 5.1.1

In each case determine whether U is a subspace of $\mathbb{R}^3$. Support your answer.

a. $U = \{(1, s, t) \mid s \text{ and } t \text{ in } \mathbb{R}\}.$

FALSE, $u \notin \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^3$

c. $U = \{(r, s, t) \mid r, s, \text{ and } t \text{ in } \mathbb{R}, -r + 3s + 2t = 0\}$

TRUE, $r, s, t \in \mathbb{R}$, ~~r can be ind.~~

where ~~ind~~ where r, s, t can be scaled to any $\mathbb{R}$, $\therefore u \in \mathbb{R}$

f. $U = \{(2r, -s^2, t) \mid r, s, \text{ and } t \text{ in } \mathbb{R}\}.$

False as has s^2 , where $s^2 \in W, s^2 \notin \mathbb{R}$ as $s^2 \in i\mathbb{R}$
 $\therefore$ does not contain vec in neg. space

explain more!

Exercise 5.1.3

In each case determine if the given vectors span $\mathbb{R}^4$. Support your answer.

- a. $\{(1, 1, 1, 1), (0, 1, 1, 1), (0, 0, 1, 1), (0, 0, 0, 1)\}$.

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \therefore \text{linear ind. 4 linear ind. rows} \Rightarrow \text{TRUE!}$$

can be scaled to rep $\mathbb{R}^4$?



Exercise 5.1.4

Is it possible that $\{(1, 2, 0), (2, 0, 3)\}$ can span the subspace $U = \{(r, s, 0) \mid r \text{ and } s \text{ in } \mathbb{R}\}$?
Defend your answer.

$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \end{bmatrix}$ FALSE;
Gives $(2, 0, 3)$ but $(r, s, 0)$
 $0 \neq 0$ $0 \neq 0$ must be!

There's no way with i, j where $U_j = 0$, to get a non-zero element out of the span of both vectors when combining both vectors.

Exercise 5.1.12

Suppose that $U = \text{span}\{x_1, x_2, \dots, x_k\}$ where each x_i is in $\mathbb{R}^n$. If A is an $m \times n$ matrix and $Ax_i = 0$ for each i , show that $Ay = 0$ for every vector y in U .

Know y must be linear combination of both vectors; $y = c_1 x_1 + \dots + c_k x_k$

Then if we wanted Ay

$\Rightarrow$ Would know can take constants out $\Rightarrow Ay = c_1(Ax_1) + \dots + c_k(Ax_k)$

$$\Rightarrow Ay = 0$$

Exercise 5.1.22

Let U and W be subspaces of $\mathbb{R}^n$. Define their intersection $U \cap W$ and their sum $U + W$ as follows: $U \cap W = \{x \in \mathbb{R}^n \mid x \text{ belongs to both } U \text{ and } W\}$. $U + W = \{x \in \mathbb{R}^n \mid x \text{ is a sum of a vector in } U \text{ and a vector in } W\}$.

a. Show that $U \cap W$ is a subspace of $\mathbb{R}^n$.

By basic set theory knowledge, it holds that

1) $U \cap W$ both have 0-vector

2) all elements in $U, W \in \mathbb{R}$

3) $U \cap W$ closed under scalar multiplication: then $cx \in U \cap W$

b. Show that $U + W$ is a subspace of $\mathbb{R}^n$.

1) Both sets 0-vector results in a 0-vector $\in \mathbb{R}^n$

2) The sum $U+W$ can still be used to generate spaces $\mathbb{R}$ for some scalars ~~can~~

3) a scalar output s/t $ca = (c(a_1 + w_1)) = ca_1 + cw_1, ca \in U + W$

Section 5.2 Independence and Dimension

Exercise 5.2.1

Which of the following subsets are independent? Support your answer.

b. $\{(1, 1, 1), (1, -1, 1), (0, 0, 1)\}$ in $\mathbb{R}^3$.

3 Reg for subsets to be ind;
 1) closed under 'x' & '+'
 2) contains 0-vector
 Then we know $\det = 0$. RREF must have 1 pivot var & all col (no free var)
 $\det \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow 1(-1+0) - 1(1-0) + 0 = -1 - 1 = -2 \neq 0 \therefore \text{subset is linearly ind.}$

d. $\{(1, 1, 0, 0), (1, 0, 1, 0), (0, 0, 1, 1), (0, 1, 0, 1)\}$ in $\mathbb{R}^4$.

$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ is one, notice: $v_1 + v_3 = (1, 1, 1, 1) = v_2 + v_4$
 $\therefore (v_1 + v_3) - (v_2 + v_4) = (0, 0, 0, 0)$ - vector
 $\therefore$ making vectors linearly independent

Test 2

Exercise 5.2.2

Let $\{x, y, z, w\}$ be an independent set in $\mathbb{R}^n$. Which of the following sets is independent? Support your answer.

b. $\{x+y, y+z, z+x\}$

1) add sm scalar to the vector/value in given set like; $\sum a(i) + b(i)$
 $A = a(x+y) + b(y+z) + c(z+x)$

2) set $A=0$ find solutions to each variable; if all $=0$, set is ind!
Then dependent set if one var $\neq 0$

~~$a(x+y)$~~

$$x(a+c) + y(a+b) + z(b+c) = 0$$

from x, y, z is ind so

$$a+c=0 \quad \therefore c=-a$$

$$a+b=0 \quad \therefore b=-a$$

$$b+c=0 \quad \text{but } (-a + -a = 0) \text{ so } a=0 \therefore c=0 \wedge b=0$$

$\therefore$ The set is independent!

To find basis 2 steps

1) Find spanning set

2) check for linear independence to throw away extra

Exercise 5.2.4

Find a basis and calculate the dimension of the following subspaces of $\mathbb{R}^4$.

b. $U = \left\{ \begin{bmatrix} a-b \\ b+c \\ b+c \\ b+c \end{bmatrix} \mid a, b, \text{ and } c \text{ in } \mathbb{R} \right\}$

To find spanning sets;
pull each var from U into own 1×4 vect
like: $\Rightarrow \begin{bmatrix} a \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -b \\ b \\ b \\ b \end{bmatrix} + \begin{bmatrix} 0 \\ c \\ c \\ c \end{bmatrix}$, factoring out vars
we have;
① $\begin{bmatrix} a \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ which represent
our spanning set
notice already $v_1 + v_3 = v_2$
so we can remove v_2
 $-a+c=b$

Basis $\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}$
dimension = 2 (no. vec in set)

e. $U = \left\{ \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \mid a+b-c+d=0 \text{ in } \mathbb{R} \right\}$

same steps,
① $a = -b + c - d$
free vars $b, c, d \in \mathbb{R}$

② $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} -b+c-d \\ b \\ c \\ d \end{bmatrix}$
-Generate span set from

$\Rightarrow \begin{bmatrix} -b \\ b \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} c \\ 0 \\ c \\ 0 \end{bmatrix} + \begin{bmatrix} -d \\ 0 \\ 0 \\ d \end{bmatrix}$

$= \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow$ none are linear
combinations
so all 3 valid

② Basis = $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

dimension = 3